Fiscal Year:	FY 2026	**Task Last Updated:**	FY 09/03/2025
PI Name:	Carr, Christopher Sc.D.		
Project Title:	Enterococci Evolution in Space: Environmental Adaptations, Antibiotic Resistance, and Clinical Implications		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	30332	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	11/04/2020	**End Date:**	11/03/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/03/2025 per F. Hernandez/ARC (Ed., 10/22/24) NOTE: End date changed to 11/03/2024 per NSSC information (Ed., 9/15/23)		
Key Personnel Changes/Previous PI:	No Co-PI/Co-I changes this year. PhD Student McKaig transitioned to a NASA FINESST award effective 9/1/22. She defended her PhD in July. Staff member Kathleen Ryan started 3/1/23 and continues working on the project.		
COI Name (Institution):	Burton, Aaron Ph.D. (NASA Johnson Space Center) Gilmore, Michael Ph.D. (Massachusetts Eye And Ear Infirmary) Wallace, Sarah Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC21K0234		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Enterococci are gram-positive bacteria that originated when our ancient animal ancestors emerged from the oceans to live on land, and brought their gut flora with them. Enterococcus faecalis (EF) and Enterococcus faecium are common human commensals and can harbor multidrug resistance. Both have been previously isolated on the International Space Station (ISS). Likely as a consequence of their evolutionary origins, enterococci show remarkable stress resistance within, but also outside, their human hosts. Their antibiotic resistance, coupled with tolerance to desiccation, starvation, and disinfection, make some EF strains potent pathogens in the built environment (e.g., hospitals), and a potential risk to crew health during space missions. The proposed study includes flight components to: 1) Characterize the frequency and genomic identity of antibiotic resistant organisms, including enterococci, on the ISS; 2) Assess the evolutionary selective pressure of the space environment (microgravity, space radiation) using EF as a model system; 3) Characterize the “natural” evolutionary history of EF on Earth and in space to reveal mechanisms of microbial adaption including natural selection. The CS-05A: Genomic Enumeration of Antibiotic Resistance in Space (GEARS) payload is designed to fulfill specific aim 1, the characterization of the frequency and genomic identity of antibiotic resistant organisms on the ISS. The Co-Principal Investigators propose to carry out longitudinal sampling of ISS surfaces in a repeated measures design. The CS-05B: Enterococcus Growth Advantage on ISS via Tn-seq (EnteroGAIT) payload is designed to fulfill specific aim 2: to assess the evolutionary selective pressure of the space environment. The flight experiment was to utilize on-board long-duration microbial growth to measure the selective pressure of the space environment on a defined microbial population: Enterococcus faecalis mutants are created by transposon insertional mutagenesis; selection is measured by sequencing (Tn-Seq) and occurs on timescales far shorter than natural or experimental evolution. The third study, Adaptation & Evolution of Resilient Enterococcus in Space (AERES) will generate complete whole genomes of Enterococcus isolates from the ISS and ground, combined with follow up characterization, to seek evidence of persistence and/or evolution in the space environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Antibiotic resistance is a growing threat to human health on Earth, resulting in infections in 2.8 million people, and causing 35,000 deaths annually (CDC data). Overuse or improper use of antibiotics is also contributing to this growing threat. Bacteria are evolving in response to the usage of antibiotics: for example, some strains of Staphylococcus aureus have acquired resistance to vancomycin from Enterococcus. Staphyloccoci and enterococci are the first and second leading causes of hospital-acquired infections, respectively. By studying the distribution of antibiotic resistant microbes on the International Space Station (ISS), a built environment similar in some ways to hospitals, we can also gain insight into how antibiotic resistant organisms survive, adapt, and evolve in response to their environment. Thus, this study will result in data that could also be relevant to human health on Earth.
	Project Acronyms (Ed., 9/10/25): Genomic Enumeration of Antibiotic Resistance in Space (GEARS) Enterococcus Growth Advantage on ISS via Tn-seq (EnteroGAIT) Adaptation & Evolution of Resilient Enterococcus in Space (AERES) Year 1 This award started on 11/4/2020. Progress to date has largely focused on project tasks required to support flight definition for CS05A (GEARS) and CS05B (EnteroGAIT), including completion of Investigative Summary Forms (ISFs) for both projects and the Science Requirements Document (SRD) for CS05B. This SRD was used by NASA to solicit proposals from implementation partners to support the CS05B investigation. A formal project kickoff was conducted on March 1, 2021, involving a NASA-PI interchange meeting with a seminar given by Principal Investigator (PI) Carr. Carr is a new faculty member at the Georgia Institute of Technology (Georgia Tech) and so initial project activities involved lab set-up for microbiology, including incubator in biosafety cabinet (for later BSL-2 Enterococcus cultures), smoke testing, certification, personal protective equipment (PPE) provisioning. Jordan McKaig, a graduate student in the Georgia Tech School of Earth and Atmospheric Sciences, joined the project. We also re-established our nanopore sequencing pipeline in the new lab (hardware, software, reagents, personnel, training). Nanopore sequencing is used in GEARS and AERES to obtain complete whole genome information from colonies and isolates, respectively. In support of the GEARS project, initial testing of the protocol to be used for in-orbit culturing and sequencing was evaluated in the lab of GEARS Co-PI Sarah Wallace (NASA Johnson Space Center). This initial data was analyzed and used to assemble two microbial genomes, simulating one potential outcome of the GEARS in-orbit culturing and sequencing experiments. We submitted an abstract to the American Society for Gravitational and Space Research (ASGSR) Annual Meeting, scheduled for November 3-6, 2021, to share initial progress. Year 2 GEARS progress: The SRD for CS05A was completed and signed in December 2021. The PI’s lab was completed in April-May 2022, and the lab was moved in preparation for Science Verification Test (SVT) and Experiment Verification Test (EVT) studies. Co-PI Wallace’s lab continued optimization of key aspects in the sample collection and preparation process. SVT studies were initiated, including an antibiotic stability and shelf-life assessment assay. Recently, the compliance review package for GEARS was developed in concert with the Mission Scientist (MS), Gwo-Shing Sun, and nominal flight candidates were identified. EnteroGAIT progress: The subaward to Massachusetts Eye and Ear (Mass Eye and Ear), Co-PI Gilmore, was initiated (initially delayed to preserve funds), the SRD was signed/approved, and BioServe was selected as the payload developer (PD) for the EnteroGAIT/CS05B hardware. A kickoff meeting was held in late July 2022. Mass estimates for the

Task Progress:	payload were made by the PD. Total crew time was estimated. The compliance review package is currently in development in cooperation with the Mission Scientist (MS). Nominal flight candidates have been identified. Initial SVT testing is underway and will include selection of the membrane material and pore size required to retain cells during media exchange, biocompatibility and growth tests, protocol verification, media stability, and optimization of sample storage, prior to a full EVT. AERES progress: This sub-project was delayed until after the PI moved to his permanent lab (May 2022). Initiation is now underway and will include nanopore sequencing of isolates from multiple sources in combination with analysis of whole genomes now available via the National Center for Biotechnology Information (NCBI). Near-term evaluation of new higher-accuracy sequencing kits (ONT Kit 14) will be performed to evaluate the potential for nanopore-only genome assembly. We submitted an abstract to the American Society for Gravitational and Space Research (ASGSR) Annual Meeting, scheduled for November 2022, to share progress on SVT work for GEARS. Year 3 GEARS Science Verification Testing (SVT) has been completed, including selection of antibiotic (gentimicin), dose, nominal growth period, and growth on contact slides with antibiotics from ISS-derived isolates from subculture and from surface swabs, followed by sequencing on MinION. Full scope testing also involved bioinformatics analysis, including genome assembly and genome-based antibiotic susceptibility assessment. EnteroGAIT: SVT work was carried out using E. faecalis OG1RF (BSL-1 strain). First, filter tests were carried out, resulting in selection of the media exchange filter of the two-chamber BioCell, under development by BioServe. The BioCells use an insert to distribute hydrostatic or hydrodynamic loads across the filter, and also to permit flexing of the outer window during loading/unloading of media or sampling of cells. The biocompatibility of the insert material was assessed using growth curves and coupons of material; this verified that either material can be used in contact with E. faecalis cells. Initial growth tests verified growth of E. faecalis in the hardware, in support of assessing validity of the nominal 8-hour post-media-addition sampling timepoint. Additional testing is planned to confirm this timepoint and to carry out longer-duration testing with the goal of retiring risk related to biofouling during long-duration culturing. Implementation Partner BioServe has already completed some long-duration culturing, e.g., nominal 8-hour growth period followed by 14 days of no new media, followed by new media and another growth and stasis period. As of early September 2023, it is anticipated that the current BioCell hardware is representative of flight and will soon be subjected to additional SVT and EVT testing (see future work, below). AERES: Oxford Nanopore Technologies (ONT) has recently developed a new kit (Kit 14) with higher accuracy. We tested this kit as part of GEARS SVT work. Based on these results, it seems likely we can simplify our AERES genome assembly work to perform nanopore-only genome assemblies, instead of hybrid genome assemblies using Illumina and nanopore data. We submitted an abstract to the American Society for Gravitational and Space Research (ASGSR) Annual Meeting, scheduled for November 2023, to share progress on SVT work for GEARS; a manuscript is in preparation. Year 4 GEARS: GEARS completed EVT on 10/26/23 and the science-oriented Mission Readiness Review (MRR) was carried out 11/9/23. GEARS launched on SpaceX CRS-30 (SpX-30) on 3/21/24, and sample collection through DNA sequencing was conducted over 4/18/24 to 5/1/24. The results of the first session are not published at this time; a summary of the results indicates that the surfaces swabbed were largely clean, with few colonies growing on the plates with antibiotics. However, multiple colonies were identified using the in-orbit sequencing workflow and the genome of the positive control was assembled based on the in-orbit data. This marks the first time successful metagenomic (all DNA) sequencing was used beyond Earth, and enhances our capability to perform research and diagnostic activities in space. EnteroGAIT: SVT work was completed in July 2024 with the EVT readiness review (EVTRR) completed on 7/24/24. Pre-EVT preparations commenced on 7/26/24, and the EVT was initiated on 8/6/24 with inoculation of BioCells with the Tn-Seq library. During this time period, Georgia Tech was without chilled water for ~3 weeks, resulting in lab conditions of 90˚F and 69% relative humidity. While all procedures went as planned, poor growth was noted in multiple BioCells, compared to SVT. The root cause was analyzed after the annual report for year 4. AERES: BSL-2 lab certification was completed in early 2024. In support of planned isolate sequencing, we recently tested Kit 14 and obtained Q20 median quality data up to 15 Gb per MinION run. We will begin sequencing isolates once EnteroGAIT is through EVT to balance project workload. Year 5 GEARS: A second session of GEARS was performed in orbit in late 2024 and into early 2025. Plates were returned and isolates generated. The first in-space generated, complete whole genomes were obtained through ground analysis of GEARS session 2 data. The first methylome generated in space was obtained through subsequent analysis of GEARS session 2 data. McKaig defended her PhD, including a chapter on GEARS (EVT-focused). An EVT-focused manuscript is under preparation. An additional No-Cost Extension (NCE) is being requested in order to support up to two additional GEARS in-orbit sessions in 2026. EnteroGAIT: The root cause of the EVT failure could not clearly be identified by the time all remaining samples were exhausted; the environmental conditions inside were notable and thought to be the cause, but no definitive result was obtained. Based on schedule and other considerations, NASA directed the project team to focus remaining resources on GEARS and AERES. AERES: Sequencing of E. faecalis isolates commenced. A new workflow for genome assembly was developed using Autocycler (https:// Application to GEARS EVT data implies near perfect genome assemblies.
Bibliography Type:	Description: (Last Updated: 12/08/2025)

Articles in Other Journals or Periodicals	McKaig JM, Carr CE. "Validation of gentamicin and streptomycin stability for on-orbit microbial monitoring." bioRxiv preprint server. Posted May 11, 2025. https://doi.org/10.1101/2025.05.11.653350 , May-2025
Articles in Peer-reviewed Journals	Bryan NC, Lebreton F, Gilmore M, Ruykun G, Zuber MT, Carr CE. "Genomic and functional characterization of Enterococcus faecalis isolates recovered from the International Space Station and their potential for pathogenicity." Front Microbiol. 2021 Jan 11;11:515319. https://doi.org/10.3389/fmicb.2020.515319 ; PMID: 33505359; PMCID: PMC7829349 , Jan-2021